# Jalen Edusei

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Software engineer who works from the hardware up: Verilog on FPGAs, embedded C/C++ on ESP32 and STM32, native macOS apps on the Claude API and on-device models, and production web apps for paying clients.

## EDUCATION

**University of Georgia**, Morehead Honors College, Athens, GA — May 2026
*Bachelor of Science in Computer Systems Engineering, cum laude* — GPA: 3.61 / 4.00

**Relevant Coursework:** Data Structures, Systems Programming, Software Development, Discrete Mathematics, Embedded Systems I & II, Advanced Digital Design, Human-Computer Interaction

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Swift, C, C++, Verilog, SQL, Elixir, Java
**Frontend:** React 19, Next.js (App Router, ISR), Tailwind CSS, three.js / React Three Fiber, Framer Motion, WCAG AA accessibility
**Backend & Data:** Node.js, Phoenix / Ecto, PostgreSQL (row-level security, transactional outbox), Supabase, REST / JSON APIs, WebSockets, SQLite / FTS5, Docker, Kubernetes, GitHub Actions, Vercel
**AI & Practices:** AI-paired development (Claude Code, Cursor), Anthropic Claude API (tool use, streaming, structured outputs), MCP servers, Agile team development, code review, automated testing (Vitest, ExUnit, XCTest)

## EXPERIENCE

**Yamaha Motor Corporation, U.S.A.**, *Full Stack Developer*, Kennesaw, GA — Nov 2026 – Present

- Maintain the AI shopping assistant on yamaha-motor.com, which finds parts, compares models, and locates dealers.

**Freelance Software Engineer**, *Self-Employed*, Atlanta, GA — May 2026 – Oct 2026

- Built cake-artista.vercel.app, a Next.js 16 storefront for a cake studio: Sanity click-to-edit for its words and 154 photo slots, a Claude assistant held to one validated tool, 1,363 tests, and mobile largest paint cut from 19.4 s on WordPress to 4.0.
- Designed, built, and shipped kulenterprises.com for a Georgia freight carrier: 22 statically generated Next.js 16 / React 19 / TypeScript pages on a git-backed TinaCMS content model, a dependency-free CMS-scoring site search, four rate-limited spam-protected forms through Resend, and WCAG AA-audited contrast.
- Extended the engagement with a broker credit-vetting portal on Supabase: a 12-question screening workflow, risk ratings with an admin-only do-not-load block, credit-exposure checks with recorded approvals, and compliance rules held in Postgres (row-level security, audit log, atomic SQL functions) behind invite-only auth with four staff roles.
- Built and shipped akilahmali.com for a recording artist: a real-time 3D control-room scene in three.js / React Three Fiber, a roughly 46 MB payload cut through code-splitting, strict security headers, and an email-capture API route.

**Capital One**, *Business Analyst Intern*, McLean, VA — Jun 2025 – Aug 2025

- Built the business case for a CreditWise Notifications Preferences Center projected to centralize communication management for 60M+ customer accounts; presented the recommendation to senior leadership.
- Analyzed email-campaign performance in SQL, Python, and Excel, creating a valuation framework that quantified engagement and retention impact and informed product-roadmap decisions.

**UGA Small Satellite Research Lab**, *Flight Software Developer (MEMESat-1)*, Athens, GA — Mar 2024 – Dec 2024

- Developed and debugged embedded C/C++ flight software on STM32 microcontrollers for the MEMESat-1 CubeSat, bringing up UART communication between subsystems and hardening it with unit and integration tests.

## PROJECTS

**Otto: AI Assistant in the MacBook Notch** | *Swift, SwiftUI, AppKit, Anthropic API* — Sep 2026

- Shipped a paid macOS assistant in the MacBook notch that streams Claude API replies over SSE with cited sources and runs calendar, Shortcuts and AppleScript actions behind hardware-only approval cards; 71,000 lines of Swift, 1,873 unit tests.

**Thataway: On-Screen Pointer for macOS** | *Swift, AppKit, Accessibility API, ScreenCaptureKit, MLX* — Aug 2026

- Built a macOS overlay that points at the control a user names, reading the accessibility tree first and falling back to a local 4-bit vision model on MLX; 12 of 12 targets on a Chrome window at 0.067 ms p50, 233 tests.
- Measured before designing: cold accessibility reads ran 6–10x slower than warm, so an `AXObserver`-driven cache became load-bearing; batching cut extraction 3.1x, and a warm ScreenCaptureKit stream beat `screencapture` 7.9 to 204 ms at p90.

**AnimalDot: Pet Vital-Signs Bed (Senior Capstone)** | *ESP32, C++, SwiftUI, Express, PostgreSQL, MQTT* — Aug 2025 – May 2026

- Led all software for a six-person capstone (36 of 37 commits) for two faculty clients: ESP32 firmware, a SwiftUI iOS app, a React Native app, and an Express/PostgreSQL backend that streams each user's vitals over WebSockets.
- Wrote real-time ballistocardiography DSP in C++ (200 Hz geophone sampling, a 0.67–3 Hz Butterworth band-pass, kurtosis-based motion rejection) and a binary MQTT telemetry protocol with tested TypeScript and Swift decoders.

**PrimeForge: FPGA Prime Finder (2-person team)** | *Verilog, Artix-7, Vivado, DDR2* — Jan 2026 – May 2026

- Designed a Verilog prime finder across four clock domains whose segmented sieve found all 5,761,455 primes below $10^8$ on hardware, streaming them to DDR2 through a two-client arbiter and gray-coded async FIFOs.

## LEADERSHIP & HONORS

Vice President, NSBE UGA Chapter (100+ members) | Tau Beta Pi | Extraordinary Engineer Award | 2x Presidential Scholar | Dean's List